

1 Prompt for Query-based Reasoning

You are given a query and a set of documents. Your goal is to reason through the documents step by step to answer the query.

Instruction:

1. Only include reasoning steps that clearly contribute to solving the query.
2. Make sure each step is logically connected and leads to the final answer.
3. Cite document snippets or facts that directly support your reasoning.
4. Avoid vague or redundant steps. Do not repeat the query.

Format:

Each step must start with [Step N]: followed by your reasoning.
 Conclude with [Answer]: followed by the final answer (and nothing else).
 Example format:
 [Step 1]: ...
 [Step 2]: ...
 ...
 [Answer]: ...

Figure 1: Prompt of query-based reasoning for teacher LLM model

2 Computation of Token-level Attention Score

In the Transformer architecture [3], the attention mechanism operates on three matrices: queries $\mathbf{Q} \in \mathbb{R}^{m \times d_k}$, keys, $\mathbf{K} \in \mathbb{R}^{n \times d_k}$, and values $\mathbf{V} \in \mathbb{R}^{n \times d_v}$. The attention function is defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}. \quad (1)$$

Here, each element $(\mathbf{Q}\mathbf{K}^\top)_{ij} = q_i \cdot k_j$ measures the similarity between the i -th query and the j -th key. Dividing by $\sqrt{d_k}$ controls the magnitude of the dot products, preventing overly large values that could destabilize training. The softmax function is applied row-wise, normalizing the scores into a probability distribution whose weights sum to one. These attention weights are then used to compute a weighted sum of the value vectors $\mathbf{V}$, producing the output representation.

At the token level, the attention weight

$$\mathbf{A}_{ij} = \text{softmax}\left(\frac{q_i \cdot k_j}{\sqrt{d_k}}\right)$$

represents how much token i attends to token j . These weights are computed dynamically for each input, enabling the model to capture semantic dependencies between tokens.

3 Proofs of Theorem 4.1 and Proposition 4.2

PROOF OF THEOREM 4.1. According to previous studies of multi-objective optimization [1], the problem of balancing competing objectives can be expressed in the weighted sum form:

$$\max_{\mathcal{D}_{\text{rerank}} \subseteq \mathcal{D}_{\text{retrieve}}, |\mathcal{D}_{\text{rerank}}| \leq K} \text{Cover}(\mathcal{D}_{\text{rerank}}) - \lambda \cdot \text{Noise}(\mathcal{D}_{\text{rerank}}),$$

where λ is a trade-off coefficient between the two objectives.

Step 1: Coverage function. From Equation (7), the coverage of the selected set is:

$$\text{Cover}(\mathcal{D}_{\text{rerank}}) = \sum_{i=1}^{|\mathcal{E}|} \mathbf{e}_i \cdot \mathbb{P}(e_i \text{ covered by } \mathcal{D}_{\text{rerank}}).$$

For each information requirement e_i , the probability it is not covered by any selected document is:

$$\prod_{d \in \mathcal{D}_{\text{rerank}}} (1 - \mathbf{w}_{d,i}).$$

Thus, the probability it is covered is:

$$1 - \prod_{d \in \mathcal{D}_{\text{rerank}}} (1 - \mathbf{w}_{d,i}).$$

Introducing the binary selection variable $\mathbf{x}_d \in \{0, 1\}$, where $\mathbf{x}_d = 1$ if document d is selected, the above product becomes:

$$\prod_{d=1}^{K_1} (1 - \mathbf{W}_{d,i} \mathbf{x}_d).$$

Therefore, the coverage term is:

$$\sum_{i=1}^{|\mathcal{E}|} \mathbf{e}_i \left[1 - \prod_{d=1}^{K_1} (1 - \mathbf{W}_{d,i} \mathbf{x}_d) \right].$$

Step 2: Noise function. From Equation (8), noise is defined as:

$$\text{Noise}(\mathcal{D}_{\text{rerank}}) = \sum_{d \in \mathcal{D}_{\text{rerank}}} \text{DisUtil}(d|q).$$

Here, disutility for a document is defined as:

$$\text{DisUtil}(d|q) = 1 - \max_i \mathbf{W}_{d,i} \mathbf{e}_i.$$

Using $\mathbf{x}_d$ to indicate selection, the noise term becomes:

$$\sum_{j=1}^{K_1} (1 - \max(\mathbf{W}_j \odot \mathbf{e})) \mathbf{x}_j,$$

where $\odot$ denotes element-wise multiplication.

Step 3: Binary optimization formulation. Combining the coverage and noise terms, and introducing the budget constraint $\sum_i \mathbf{x}_i \leq K$, we obtain:

$$\begin{aligned} \max_{\mathbf{x}} \quad & \sum_{i=1}^{|\mathcal{E}|} \mathbf{e}_i \left[1 - \prod_{d=1}^{K_1} (1 - \mathbf{W}_{d,i} \mathbf{x}_d) \right] - \lambda \sum_{j=1}^{K_1} (1 - \max(\mathbf{W}_j \odot \mathbf{e})) \mathbf{x}_j, \\ \text{s.t.} \quad & \sum_i \mathbf{x}_i \leq K, \quad \mathbf{x}_i \in \{0, 1\}, \forall i. \end{aligned}$$

This exactly matches the statement in Equation (9), proving that the denoised coverage-aware document selection problem can indeed be formulated as a 0–1 integer (binary) multi-objective optimization problem. $\square$

PROOF OF PROPOSITION 4.2. We prove the claim by showing each per-evidence term is monotone and submodular, and then use closure properties of submodular functions. Let's define the multi-objective objective function in Problem (6) as $g(S) = G(S) - \lambda \cdot R(S)$.

(1) Per-evidence term. Fix an evidence e_i . Define

$$h_i(S) = 1 - \prod_{d \in S} (1 - \mathbf{W}_{d,i}), \quad S \subseteq \mathcal{D}.$$

For any $S \subseteq T \subseteq \mathcal{D}$ and any $d \in \mathcal{D} \setminus T$ the marginal gain of adding d is

$$\Delta_d(S) = h_i(S \cup \{d\}) - h_i(S) = \prod_{t \in S} (1 - \mathbf{W}_{t,i}) - \prod_{t \in S} (1 - \mathbf{W}_{t,i})(1 - \mathbf{W}_{d,i}) = \mathbf{W}_{d,i} \prod_{t \in S} (1 - \mathbf{W}_{t,i}).$$

Similarly,

$$\Delta_d(T) = \mathbf{W}_{d,i} \prod_{t \in T} (1 - \mathbf{W}_{t,i}).$$

Since $0 \leq 1 - \mathbf{W}_{t,i} \leq 1$ for all t , we have

$$\prod_{t \in S} (1 - \mathbf{W}_{t,i}) \geq \prod_{t \in T} (1 - \mathbf{W}_{t,i}),$$

and therefore $\Delta_d(S) \geq \Delta_d(T)$. This is exactly the diminishing returns property, so $h_i(\cdot)$ is submodular. Moreover $\Delta_d(S) = \mathbf{W}_{d,i} \prod_{t \in S} (1 - \mathbf{W}_{t,i}) \geq 0$, so $h_i(\cdot)$ is monotone non-decreasing.

(2) Sum preserves submodularity and monotonicity for Coverage Objective. The coverage objective $G(S) = \sum_i \mathbf{e}_i h_i(S)$ is a nonnegative linear combination of the functions h_i . Nonnegative linear combinations of submodular (resp. monotone) functions remain submodular (resp. monotone) [2]. Hence $G(\cdot)$ is monotone and submodular.

(3) Per-document term. Fix a document d . Define its disutility score as

$$\text{DisUtil}(d) = 1 - \max_{e \in \mathcal{E}} \mathbb{P}(d | e) \mathbb{P}(e | q).$$

Now consider the noise function over a set of documents

$$S \subseteq \mathcal{D} : R(S) = \sum_{d \in S} \text{DisUtil}(d).$$

Notice that each per-document term $\mathbf{c}_d \cdot \text{DisUtil}(d)$ depends only on d and the evidence set $\mathcal{E}$, but not on the other documents in S . Hence, each term contributes additively and independently.

(4) Submodularity and monotonicity of Noise Objective. For any $S \subseteq T \subseteq \mathcal{D}$ and any document $d \in \mathcal{D} \setminus T$, the marginal gain of adding d is

$$\Delta_d(S) = R(S \cup \{d\}) - R(S) = \text{DisUtil}(d).$$

Similarly,

$$\Delta_d(T) = \text{DisUtil}(d).$$

Since the marginal contribution of d is identical regardless of the context set, the diminishing returns property holds trivially. This means $R(\cdot)$ is modular (a special case of submodular) [2].

Moreover, as $\text{DisUtil}(d) \geq 0$, we have $\Delta_d(S) \geq 0$, implying monotonicity. Therefore, the noise objective

$$R(S) = \sum_{d \in S} \left(1 - \max_{e \in \mathcal{E}} \mathbb{P}(d \mid e) \mathbb{P}(e \mid q) \right)$$

is a monotone modular function, and hence also submodular.

(3) NP-hardness. Maximizing our proposed objective $g(\cdot)$ under a cardinality constraint is generally NP-hard. A cardinality constraint means that the solution set S is restricted to contain at most K documents (i.e., $|S| \leq K$), which reflects the practical setting of re-ranking where we can only select a limited number of documents for the generator.

This complexity includes the well-known Max-k-Cover problem as a special case. Specifically, if each document d either fully covers a particular information requirement e_i ($p_{d,i} = 1$) or does not cover it at all ($p_{d,i} = 0$), then the coverage function $h_i(S)$ becomes an indicator function that equals 1 if e_i is covered by at least one document in S , and 0 otherwise. Under this setting, maximizing $g(S)$ is equivalent to selecting K documents to cover as many weighted information requirements as possible, which is exactly the Max-k-Cover problem, which is known to be NP-hard [2].

Therefore, even in the more general case where $p_{d,i}$ takes continuous values (representing partial coverage), maximizing $g(\cdot)$ remains NP-hard. This justifies the need for an efficient approximation algorithm, such as our proposed greedy approach. $\square$

References

- [1] Kalyanmoy Deb, Karthik Sindhya, and Jussi Hakanen. 2016. Multi-objective optimization. In *Decision sciences*. CRC Press, 161–200.
- [2] Rishabh Krishnan Iyer. 2015. *Submodular optimization and machine learning: Theoretical results, unifying and scalable algorithms, and applications*. Ph.D. Dissertation.
- [3] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).